

**Workers Without Work:
Pigeons and the Infrastructural Afterlife of Domestication**

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Abstract

In January 1944 the Office of Scientific Research and Development declined to extend the contract on a missile guidance system built out of pigeons, and B. F. Skinner recorded the two outcomes of that decision in one sentence: the equipment was scrapped, and 30 of the birds were kept to see how long they would hold the behaviour. They held it for six years. This paper is about the general shape of that sentence. Rock doves entered human settlements by the commensal pathway and were enrolled, over roughly five millennia, as meat, as a manure engine, as a message-carrying network, as military signals equipment, as a laboratory model, as a racing athlete, and as a religious sign. The enrolments ended asynchronously, decided by unrelated institutions that each withdrew a function without any of them ending the population, a process this paper calls distributed decommissioning. Its central result is that what a decommissioned working-animal population becomes is fixed by two institutional variables rather than by the animal's traits: whether the population stays enumerable, each individual still attached to an owner and so reachable by a decision, and whether any institution keeps a standing obligation to it. All four combinations occur in the record, and three of them are written into statute. Military working dogs pass at the end of their useful working life into adoption; research chimpanzees pass into a sanctuary system that may never discharge them; unbranded and unclaimed horses on the public range were declared living symbols and protected from capture and death, at a cost now visible in a range population above 85,000 against a management level of 25,592. The Army's own pigeons, being listed, ended cleanly: 15 to zoos, about 1,000 sold, the last bird auctioned on 23 March 1957. The urban horse, owned to the last day and worth 24 dollars as a carcass, went down the market and through the rendering works and left no feral population behind it. The street pigeon was reachable by no decision and worth nothing dead, and the standard wildlife-damage handbook states its position in a line: feral pigeons hold no federal protection. What follows from deownership is a reversal of valuation that the control literature documents against itself, since the reason it gives for the impracticality of release is homing, the capacity fanciers spent centuries selecting for. The paper then reads pest status as a relation among recognized utility, ownership, and spatial authorization, restores each term in turn to watch the category dissolve, traces the canonical billion-dollar damage figure back to an assumed cost per bird multiplied by an assumed bird per hectare, and closes on the spaces these birds occupy, which are habitable in fact and disavowed as habitat, held that way by a continuous expenditure on spikes patented in 1939 and sold commercially since 1950. A society can retire a function faster than it can retire the population that performed it.

1. The Equipment Was Scrapped

Skinner's account of Project Pigeon is unusually candid about how an institution ends a relationship with an animal. The system worked. A bird in a harness, pecking at an image of a target projected on a screen at the nose of a missile, steered it; three birds voting steered it better; and the demonstration before the committee of physical scientists was, in Skinner's words, a perfect performance that had just the wrong effect, because the sight of a living pigeon carrying out its assignment reminded the committee how fantastic the whole proposal was (Skinner, 1960). The reason

given for refusing to extend the contract was that other projects had more immediate promise of combat application. The equipment was scrapped. The 30 retained birds became an experiment in the persistence of a capacity after the disappearance of its use, and Skinner reports the result plainly: even after six years of inactivity a pigeon will immediately and correctly strike a target to which it has been conditioned.

That is the situation this paper generalizes. A working machine and a working animal both stop being wanted, and only one of them stops. A machine's decommissioning is complete at the moment of the decision, because a decommissioned machine has no further requirements, makes no further claims, and produces no further machines. An animal's decommissioning is the beginning of a second history, in which a competent, hungry, reproducing population goes on existing inside an institutional order that has stopped having a place for it. Skinner's 30 birds had somewhere to be, because they were on a list and someone was measuring them. The general case is a population that is on no list.

Feral urban pigeons are the largest and longest-running instance of the general case. Their ancestors were enrolled in more distinct human systems than almost any other animal, and those systems were dismantled at different times, by different actors, for different reasons, across about two centuries. Nobody ever declared the birds abandoned. The declaration was never needed, because the withdrawal happened one institution at a time, and no institution had ever held the whole population to begin with. The result is an animal that is simultaneously a human artefact and nobody's responsibility, present in tens of millions across the world's cities, and described by the discipline that manages it as the most common bird pest associated with people (Williams and Corrigan, 1994).

The argument runs in five moves. Section 2 reconstructs what the bird was for, as a set of institutions rather than a set of uses. Section 3 defines distributed decommissioning and separates it from abandonment. Section 4 is the central result, a two-variable account of what becomes of a decommissioned animal population, with all four outcomes instantiated and three of them enacted in law. Section 5 shows the inversion of the animal's capacities as a documented feature of the control literature. Sections 6 and 7 take the resulting category apart, first as a relation among three institutional terms and then as a claim about space.

2. What the Bird Was For

Columba livia came in through the commensal pathway, the one that begins without any human intention to domesticate. People built, and animals attracted to the built environment stayed (Zeder, 2012; Larson and Fuller, 2014). Cliff-nesting birds found masonry ledges and cavities adequate, and a species that is an obligate synanthrope today became one by moving out of rocky habitat into permanent towns in the Near East. The archaeological record of deliberate management now reaches back further than it did a decade ago. At Hala Sultan Tekke in Late Bronze Age Cyprus, rock doves were breeding inside the settlement by about 1400 BCE, their stable isotope values overlapping the human dietary range in a way that indicates provisioning with grain, and their bones carrying the burning patterns of birds that were eaten, including in the cultic spaces where feasting

took place (Carter et al., 2026).

The first large institution built on the bird was agricultural, and its product was excrement. A dovecote is a machine for converting unsupervised foraging into private fertility: the birds feed themselves at large and return to a structure whose owner harvests the squabs and the manure. The scale is legible in the excavated cases. At Byzantine Sa'adon in the Negev, a circular dovecote 6.2 metres across held roughly 1,100 nesting cavities and therefore about 550 pairs, producing an estimated 6,000 to 7,000 hatchlings and 10 to 11 cubic metres of manure a year; a smaller square structure nearby held about 600 compartments and 300 pairs (Tepper et al., 2018). The excavators found the second building's floor to be a layer of pigeon dung with articulated skeletons in it, and the first swept clean, its manure presumably removed in an orderly fashion before the site was abandoned. Mishnaic sources prescribe a dose, 500 litres of pigeon manure per 1.24 acres per year. In Iran the same institution reached industrial scale. The towers of the Isfahan region were standardized down to nesting holes of 20 by 20 centimetres and 27 deep, they were entered once a year, and seventeenth-century travellers counted more than 3,000 of them, one of which was said to hold 10,000 birds and to provide the chief income of its village. Beazley, surveying the survivors in the 1960s, is explicit that the function of the towers is the collection of manure, and that both towers and birds belonged to the landlord, who paid the Shah a tax on the manure he sold (Beazley, 1966).

The second institution was communication, and it was built on homing. A pigeon post is never a bird with a message attached. It is a loft, a breeding programme, handlers, an arrangement for carrying birds away from the loft, and a route, with the animal supplying one component of a distributed addressing system. Julius Reuter's service between Aachen and Brussels, which moved market prices across the gap between the Prussian and Belgian telegraph networks from 1850, is the cleanest miniature of the whole history, because its end is dated: the wire closed the gap and the service closed with it in 1851 (Read, 1992).

The third was military, and it was the largest. The United States Army's pigeon service was authorized in November 1917 and disbanded forty years later. In the Second World War the Signal Corps furnished about 54,000 birds to the armed services, of which roughly 40,000 were donated by civilian fanciers, which is to say that a state communications capability was assembled out of a hobby. On the British side, Operation Columba dropped more than 16,000 pigeons into occupied Europe between April 1941 and September 1944, and about one in ten came back, some 1,000 of them carrying messages (Corera, 2018). Of the 75 Dickin Medals awarded since 1943, 32 have gone to pigeons, more than to any animal except the dog. The individuation was administrative before it was affective, and the archive shows the machinery working: Blazich's reconstruction of Cher Ami finds nothing conclusive linking the bird to the action it is famous for, and concludes that the Army chose to link the pigeon to the Lost Battalion's story in order to promote the Signal Corps' Pigeon Service (Blazich, 2021). The most individuated animal of the twentieth century was a public relations decision, and its sex was recorded wrongly for a century.

The fourth was scientific. Darwin gave two chapters of *Variation* to pigeons and said why: one case fully described would illustrate all the others, and the materials for this one were better than for any

other (Darwin, 1868). His argument turned on the taxonomist's eye. At least a score of the breeds, shown to an ornithologist and described as wild, would be ranked as well-defined species, and all of them descend from *Columba livia*. Selection was thereby shown sufficient to produce divergence of a magnitude that looks generic, which is the analogy the theory of natural selection required. Skinner's use eighty years later was of the same kind, treating the animal as an instrument whose reliability was the point.

The fifth was social, and it is the one that survived longest. Jerolmack's ethnography of the rooftop flyers of New York describes working-class men whose birds organize their perception of the neighbourhood, and whose competitive gatherings at a pet shop produce status out of flight performance, generating solidarity across Italian, Hispanic and African American lines where the community-studies literature would predict conflict (Jerolmack, 2009). The birds were producing something here too, and it was neither protein nor information.

Read together, these are five institutions with an animal inside them rather than five uses of an animal, which is the sense in which pigeons belong to the study of infrastructure rather than to the study of pets: non-human life recruited as infrastructure, made to circulate information or fertility on terms set by an organization (Barua, 2021). The work was work in the strict sense the animal-labour literature has been arguing for, done by the animal rather than merely with it or for it (Hribal, 2003; Coulter, 2016).

3. Distributed Decommissioning

Call the piecemeal withdrawal of functions, provisioning, and responsibility from a living human-made population *distributed decommissioning*. The term is available; the engineering literatures that own the word apply it to reactors, rigs, and hardware, and none of them has had to consider a decommissioned system that eats.

The withdrawal was distributed along three axes. It was distributed in time, since the food and manure regimes, the message networks, the military services, and the laboratory uses ended across roughly two centuries, and the sporting and religious uses have still not ended. It was distributed across institutions, since armies, news agencies, agricultural estates, municipalities, households, and clubs each decided only about their own birds. And it was distributed in space, since the leakage into cities came from lofts, ports, farms, race routes, and military sites everywhere at once, so that urban populations accumulated rather than issuing from any release event.

The consequence is that the transition required no decision by anyone. A service closed. A dove-cote stopped being maintained. A race bird failed to return. A household stopped keeping birds. Ferality, in this account, is the residue of an omission, and the genetic record shows the omission to be continuous rather than historical: racing breeds have made substantial contributions to feral populations, and feral and racing birds are only minimally differentiated (Stringham et al., 2012; Gilbert and Shapiro, 2014). The homing faculty that made the birds useful is the same faculty that let owners fly them free by day, which is what supplied the opportunity for them to go feral in the first place. Hybridization has run so far in the other direction that genuinely wild rock doves may

be approaching genotypic extinction, and the principal threat to the wild form is now replacement by ferals (Hernández-Alonso et al., 2023).

Ferality of this kind is not a return journey. The feralization literature is emphatic that domestication-related changes are not predictably reversed, that retained domestication alleles persist and are sometimes adaptive in the new environment, and that where an ancestral trait does reappear it often arrives by a novel genetic route rather than by rewinding the original one (Gering et al., 2019). A feral pigeon is a domestic animal whose owner has been subtracted. Its body, its behaviour, its tolerance of human proximity, and its geographic distribution remain consequences of human history, and the city is the only environment for which it was ever fitted.

One case in the record shows what the alternative looks like, and it involves the same species. When the Army disbanded its pigeon service, the withdrawal was a dated administrative act with an inventory attached. Fifteen surviving decorated birds went to zoos, about 1,000 were sold to the public, and the last bird was auctioned on 23 March 1957. That decommissioning ended, because the Army knew how many pigeons it had and where each of them was. No such sentence can be written about the street.

The final enrolment is being withdrawn now, and it is counting itself as it goes. Britain's Royal Pigeon Racing Association put its own membership at roughly 60,000 in 1989 and at just under 14,000 in November 2024 (Royal Pigeon Racing Association, 2024). What is disappearing there is the last institution that held individual birds to individual names, and its disappearance removes both a source of escapes and the only body of people who could identify a particular pigeon.

Edgerton's use-centred history is the right corrective to the replacement narrative that would otherwise organize this material (Edgerton, 2007). The telegraph did not abolish the pigeon post; it made one particular pigeon post unprofitable in 1851, while military pigeon services expanded for another century. Chickens did not abolish squab. What happened was slower, more local, and more institutional than the word obsolescence suggests, and the population never received the signal at all.

4. Why the Horse Left and the Pigeon Stayed

The comparison that disciplines this argument is the urban horse, because the horse was decommissioned faster and more completely than the pigeon and left no feral urban population behind it.

The scale was enormous. Manhattan held about 130,000 horses in 1900, Chicago 74,000, Philadelphia 51,000, roughly one animal for every 20 people in large nineteenth-century cities, and the national horse and mule population peaked in the 1910s at around 26.5 million before falling to about 3 million by 1960 (McShane and Tarr, 2007; Kilby, 2007). The manure economy was of the same order as the dovecote's and met the same end: a heavy urban animal produced between 30 and 50 pounds a day, some seven tons a year, and the market for it was destroyed by mineral phosphates.

What matters here is the exit. The urban horse moved down a market. It was broken to harness on a farm, worked by a street railway that depreciated it over five years, sold to a delivery firm, then to a cab firm, then to a peddler, with the oldest and least sound animals at the bottom of the chain. McShane and Tarr summarize the end of that career in one sentence: lameness and then the veterinarian's pistol was the most common end for city horses. They also give the economic reason the system closed, which is that the carcass had a market. A dead horse went to a rendering plant, the organic equivalent of a scrap iron processor, and yielded hair, leather, hoof oil, bone, phosphorus, pet food, dyes, and soap; one St. Louis firm in 1896 claimed a profit of 24 dollars per carcass. New York's rendering monopoly guaranteed the Board of Health removal within three hours, because hides lost value once rigor mortis set in, and in 1910, probably the peak year, the city reported taking more than 20,595 dead horses off its streets. The killing was industrial and much of it was done by the humane society: between 1887 and 1897 New York's ASPCA shot between 1,800 and 7,000 horses a year, which McShane and Tarr call the beginning of the paradox by which humane societies became the leading killers of animals in cities.

The urban horse therefore had two properties the street pigeon lacks. It stayed *enumerable*, meaning that every individual was attached to an owner and could be reached by a decision. And it had a positive terminal value, which gave the last owner a reason to complete the transaction rather than release the animal. Nothing about horses makes them incapable of going feral; the mustang exists. What made the urban horse's disappearance total is that the institution kept an address for every animal until the end.

Two variables organize the outcomes. The first is enumerability. The second is whether any institution retains a standing obligation to the population after the work stops. Crossing them gives four terminal states, and each of the four occurs.

	Obligation retained	Obligation withdrawn
Enumerable	Retirement: military working dogs adopted at the end of their useful working life; research chimpanzees moved to a sanctuary that may not discharge them	Liquidation: the urban horse sold down the market and rendered; the Army's own pigeons sold off in 1957
Not enumerable	Reassumption: unbranded and unclaimed horses and burros on the public range, protected by statute and managed in perpetuity	Pestification: the feral pigeon

Three of these cells are written into law, and reading the statutes side by side is the fastest way to see that the pigeon's status was settled by default rather than by anything about pigeons. The military working dog provision, enacted in 2000, is titled "Military working dogs: transfer and

adoption at end of useful working life”, and it exists because until that year the standard disposition of a dog whose working life had ended was euthanasia (United States Congress, 2000a). The chimpanzee statute of the same year established a sanctuary system for animals designated as no longer needed in research, obliged it to provide lifetime care, and prohibited the chimpanzees from being discharged from the system (United States Congress, 2000b). The 1971 horse statute is the most instructive of the three, because it legislates directly for the unenumerable case. It defines its protected class as all unbranded and unclaimed horses and burros on public lands, declares them living symbols of the historic and pioneer spirit of the West, protects them from capture, branding, harassment, or death, and requires that they be considered, in the area where they are found, an integral part of the natural system of those lands (United States Congress, 1971). A population with no owner was given a legal residency and a standing claim on public money.

The price of that decision is on the books. As of March 2026 the Bureau of Land Management put the on-range population at 85,466 horses and burros against a national appropriate management level of 25,592, and the difference is carried by a holding programme that keeps tens of thousands of removed animals in corrals and pastures for the rest of their lives (Bureau of Land Management, 2026). The National Research Council’s review found the programme had not used scientifically rigorous methods to estimate population sizes or to model the effects of its own management actions (National Research Council, 2013). Reassumption is expensive and it is badly measured. It is also the only cell in the table where a decommissioned unowned population is treated as having a claim.

Against those three statutes, set the line the wildlife-damage handbook prints under the heading Legal Status: feral pigeons are not protected by federal law and most states do not afford them protection (Williams and Corrigan, 1994). The same document opens its account of the animal by naming its origin. The common pigeon was introduced to the United States as a domesticated bird, many escaped and formed feral populations, and the pigeon is now the most common bird pest associated with people. The causal story and the administrative verdict sit two pages apart, and nothing in the document connects them.

5. The Capacities Inverted

Deownership does more than remove a status. It reverses the sign on the animal’s traits, and the reversal is legible because the same capacities appear in the husbandry literature as achievements and in the control literature as obstacles.

Capacity	Valued as	Encountered as
Homing and site fidelity	The basis of the message network and the race	The reason release is impractical, since birds return from 80 kilometres away
Tolerance of human proximity	Ease of husbandry, the condition of the loft	Feeding by the public, an urban food base to be suppressed

Capacity	Valued as	Encountered as
Flocking	Efficient management and collective release	Infestation, and a distribution channel for flock-alarming poison
Fecundity	Renewable squab, a fresh clutch before the last has fledged	Overpopulation and the failure of removal
Ledge and cavity nesting	Compatibility with the dovecote and the loft	Fouling of buildings, to be engineered away at 45 degrees
Droppings	High-quality manure, taxed and sold	Corrosion, contamination, and a cleaning bill

The strongest evidence for the inversion is that the control literature explains its own difficulties by naming the products of domestication. Under the heading of trapping, the handbook considers what to do with birds once they are caught, and rules out the humane-seeming option in two sentences: releasing pigeons back to the “wild” is impractical, because pigeons are likely to return even when released 50 or more miles from the problem site, or become pests in other communities (Williams and Corrigan, 1994). The scare quotes are the handbook’s own. There is no wild for this bird to be released into, and the faculty that prevents its release is the faculty fanciers selected for across centuries and armies paid for. The recommended alternative is a carbon dioxide chamber built from a modified garbage can or a 55-gallon drum, in which birds expire in five to seven minutes.

The document performs the inversion elsewhere with less self-awareness. Its list of pigeon habitat opens with lofts, steeples, and attics, so the husbandry structure appears first in an inventory of pest sites. Its account of the food base observes, in passing, that in some urban areas the feeding of pigeons is considered a form of recreation, and then recommends discouraging people from feeding pigeons in public areas, which is a measure aimed at humans. Its use of Avitrol converts gregariousness into a delivery mechanism, since the dying birds exhibit distress behaviour that frightens the rest of the flock away, and the manual advises diluting the bait to maximize the alarming reaction while minimizing visible mortality in places where a public might object. Sociality is the delivery system.

The inversion is a fact about the institution’s relation to the animal rather than a discovery about the animal. Between the loft and the ledge the bird’s capacities hold constant, and the sign flips.

6. Three Terms and a Classification

Pest status is a relation among an organism, a recognized utility, an ownership claim, and an authorized place. Restore any one of the three institutional terms and the classification dissolves without a single change to the bird. A banded bird in a loft is an athlete. A bird in a laboratory is

a model organism. A bird fed in a temple courtyard is a sacred presence. On a ventilation unit it is vermin.

Jerolmack dates the assembly of the category. Reading the *New York Times* from 1851 and 51 other papers from 1980, he finds pigeons treated as an occasion for complaint but not as a species-level problem until the mid-1930s, when a letter-writer first says that pigeons outside parks and squares are entirely out of place. Disease enters in 1945, consolidates through the 1950s, and turns lethal in the reporting of 1 October 1963, when a city health official ascribed two deaths to pigeon-borne disease and called for a campaign against the city's 5 million pigeons, the first occasion on which a population estimate was published alongside a body count. The phrase that fixed the category arrived three years later, on 22 June 1966, when the parks commissioner Thomas Hoving called the pigeon a rat with wings; the line reached the public through a mayor quoting it approvingly in 1979 and a Woody Allen film in 1980 that is often mistaken for its origin (Jerolmack, 2008). The sequence runs from spatial complaint to medical frame to extermination programme, and the medical frame arrives after the spatial one.

Jerolmack reads this through Douglas, and the reading is right, though the provenance of the borrowed phrase is better than either of them notices. Douglas presents "matter out of place" as the old definition of dirt, inherited rather than coined, and the earliest documented instance is Lord Palmerston addressing the Royal Agricultural Society in 1852: he had heard it said that dirt is nothing but a thing in a wrong place, and the dirt of our towns corresponds to that definition (Campkin, 2013; Fardon, 2013). The context of that sentence was a debate about whether urban waste could be carried back to the land as fertilizer. The formula that now condemns pigeon droppings on a monument was minted in an argument about how to get excrement back into agriculture, which is the argument the dovecote had already settled thirteen centuries earlier. Inside the tower the droppings were the product, taxed and sold. On the cornice they are the offence. The chemistry is unchanged, and the infrastructure of collection, ownership, and authorized use is gone (Douglas, 1966).

The health claims are real and small. The most careful review of the literature found 176 documented transmissions of illness from feral pigeons to humans between 1941 and 2003; the birds harbour some 60 organisms pathogenic to humans, of which 7 were actually transmitted, and 99.4 per cent of the documented incidents were aerosol transmissions, which is why heavy cleaning of accumulated droppings is the exposure that matters and why immunocompromised people carry most of the risk (Haag-Wackernagel and Moch, 2004). New York City's health guidance says as much in plain language: the risk of pigeon-related diseases is rare, those most at risk have compromised immune systems, routine cleaning of a windowsill does not pose a serious health risk to most people, and it is not illegal to feed or keep pigeons. A hazard of that size is doing a great deal of classificatory work, and the work it does is spatial.

The economic claims deserve the same treatment, and they have a traceable origin. The figure of 1.1 billion dollars a year in United States pigeon damage, which now circulates in trade literature without attribution, comes from a single paragraph in an invasive-species survey, where it is produced by taking a control cost of 9 dollars per pigeon per year, assuming a density of one pigeon

per urban hectare, and using potential control costs as a surrogate for losses (Pimentel et al., 2000). The assumed inputs are an estimate and a stipulation, the arithmetic is a multiplication, and the same passage is reprinted unrevised five years later. The methodological critique of that literature is now published (Boltovskoy et al., 2022). The number is not a measurement of anything, and it has been doing the work of one for a quarter of a century.

Density itself is an artefact of the city's food regime rather than a property of the species, and two natural experiments show it. In a controlled trial across two Barcelona districts, a public campaign to reduce deliberate feeding cut pigeon numbers by 40 per cent between February and June while the control district did not vary, and the detailed analysis attributed the fall to reduced food availability rather than to the culling that ran alongside it, even though the culling removed more than 12,000 birds (Senar et al., 2017). In Venice, where an estimated 350 tons of corn a year were being fed to birds concentrated at more than 10,000 individuals on the 1.3 hectares of St Mark's Square, the authorities banned the sale and distribution of corn in May 2008, and counts fell to a maximum of about 1,000 birds at a time (Giunchi et al., 2012). Both interventions worked on the humans. Neither worked on the birds.

Thompson's rubbish theory supplies the last piece and then breaks on it. His scheme has three categories, transient, rubbish, and durable, with movement permitted from transient to rubbish and from rubbish to durable and blocked in the other directions; the rubbish category is covert, a region of zero value where objects wait to be rediscovered (Thompson, 1979). A decommissioned animal enters that region and refuses to behave, because rubbish that metabolizes does not wait. It feeds, it moves, it selects new sites, and it produces more of itself, so the covert category fills rather than sitting still, and the classification has to be maintained by expenditure rather than by neglect.

The word for that maintenance is available with a caveat. *Pestification* already exists in plant pathology, where it names the evolutionary adaptation of a pathogen from a wild host to crop conditions, a process occurring in the organism (Saleh et al., 2014; De Gracia et al., 2015). The process at issue here is the administrative counterpart: a change in the relation rather than in the animal, combining lost utility, withdrawn ownership, deindividualization, sanitary framing, and spatial illegitimacy. Both senses concern domestication producing a pest. Only one of them requires the pest to change.

7. Feral Nonspace

Augé's non-place is defined by a relation and not by a site. A place is relational, historical, and concerned with identity; a space that cannot be so defined is a non-place; and the two are opposed polarities rather than physical kinds, palimpsests on which identity and relations are ceaselessly rewritten, so that the non-place never exists in pure form (Augé, 1995). The concept was built for the human user of the motorway, the airport, and the supermarket. It says nothing about whether the same concrete is relational, historical, and identity-bearing for something else standing on it.

For a pigeon, a station concourse is a cliff with a food supply. It is the site of courtship, nesting, flock association, route learning, and intergenerational occupation, which is the full list of things Augé requires of a place. The commuter's non-place and the bird's territory are the same building.

Star's account of infrastructure already contains the licence for this move, since her properties are explicitly relational and her own example is a staircase that is infrastructure for one body and a barrier for another (Star, 1999). Extending the relativity across species changes nothing in either framework and produces a category that cities spend money to deny.

Pigeons hold an unusual position with respect to infrastructure, having occupied both sides of it. They were components of communication systems, and they now inhabit the physical systems that replaced those. Barua's account of non-human life and infrastructure names the second relation directly: infrastructures become a medium of inhabitation through repurposing, when animals use them against their design intent (Barua, 2021). Contemporary ethnography shows what the inhabitation looks like. Haque's study of Karachi describes interstitial ecologies formed in road margins and overlooked in-between spaces, where cohabitation is shaped by class, belief, and everyday feeding practice (Haque, 2026). Juola's account of a pigeon colony in an abandoned pulp-mill silo in Oulu shows the birds thriving in a structure people no longer consider valuable (Juola, 2024).

Star's best-known claim is that infrastructure becomes visible upon breakdown. The pigeon inverts it. This infrastructure became visible when it stopped working, invisible while it carried messages and manure inside functioning institutions, hypervisible once the institutions withdrew and the birds remained on the outside of the buildings. Visibility here is a symptom of decommissioning rather than of failure.

Call anthropogenic space that is materially habitable and institutionally disavowed as habitat *feral nonspace*. Bridge undersides, ventilation cavities, station canopies, cornices, and industrial ruins are its instances. The disavowal is expensive, and the expense is the evidence. Spike-type bird-proofing was patented in 1939, and the porcupine wire familiar from every ledge has been sold commercially since 1950 (Peles, 1939; Burnside et al., 1957). The handbook's exclusion measures amount to a programme of architectural revision: mesh the vents, net the rafters, string monofilament at three heights, change the angle of any roosting ledge to 45 degrees or more. A building that requires all of this in order to stay a building rather than a cliff was never as unavailable to the birds as its designers assumed, and a ledge that has to be re-angled is a habitat with a countermeasure on it.

The comparison with hostile architecture aimed at people is available and has to be handled carefully. The design vocabulary is chronologically prior on the animal side, since the anti-bird spike precedes the anti-homeless spike by decades, and when the London spikes controversy broke in 2014 the public reached for the comparison unprompted, objectors observing that the devices resembled the mechanisms used to keep pigeons off ledges and one resident saying that it was like treating people like animals (Petty, 2016). The transferable claim concerns technique. A city that has learned to make a surface uninhabitable has a method, and methods travel. The subjects do not become equivalent because the hardware does, and the histories, rights, and vulnerabilities involved are not interchangeable.

What the birds do to these spaces is why the spending never ends. Philo and Wilbert distinguish

the animal spaces that human orderings allot from the beastly places animals make for themselves, and observe that animals evade the places to which humans seek to allot them (Philo and Wilbert, 2000). Repeated return, nesting, and occupation across generations convert a surface into a place, so the emptiness of the non-place is a claim that has to be enforced continuously against something that is trying to live there.

8. What Is Owed to a Population Nobody Ordered

The obligation question is unavoidable once the mechanism is stated, because the mechanism identifies a population that exists as a consequence of human action and has no counterpart institution facing it.

Donaldson and Kymlicka's category fits the case, and they name pigeons in it. Liminal animals are neither domesticated co-citizens nor sovereign wildlife; they live among us without being ours, and the status assigned to them is denizenship, with three components. Secure residency means the animals are entitled to stay rather than to be treated as invaders. Fair terms of reciprocity permit humans to regulate shared space and to limit reproduction by means that do not violate the animals' basic rights. Anti-stigma safeguards protect them from vilification as vermin, which is the term the control literature uses (Donaldson and Kymlicka, 2011). Within the animal-labour literature the same conclusion arrives from the direction of work: recognition as a member of a society carries entitlements including retirement provision, and the animals to whom societies have granted anything of the kind are the police and military animals whose quasi-membership was already established (Cochrane, 2019).

The instrument that matches the diagnosis is the supervised loft. A municipal dovecote restores all three of the terms whose absence produces the pest. It names a responsible party, which is ownership. It designates a location, which is spatial authorization. And it produces something, which is utility, in the form of collectable droppings, observable health, and controllable reproduction. The German schemes have run on this model for three decades. The honest report on them is that the mechanism is sound and the evidence is thin: the specialist review of pigeon control observes that it is not aware of any quantitative estimation of the efficacy of urban dovecotes with egg replacement, and says the same about culling, which has never been shown to affect population size (Giunchi et al., 2012). What has been demonstrated is the food base, twice, in Barcelona and in Venice. A regime built on that finding would manage the resource and the sites rather than pursuing total exclusion, and its point would be to make the birds addressable again.

None of this makes the birds harmless or the complaints imaginary. Droppings corrode stone, cleaning is expensive, and a person clearing a heavy accumulation without protection faces a real if small risk. Nor does the argument romanticize adaptability. Urban pigeons are short-lived, and a population cut by roughly 30 per cent a year by predation, disease, hunger, and control is not flourishing in any sense that would satisfy the people who claim to admire its resilience.

The general form of the problem outlives its instance. A technical system can withdraw a function in a decade, and the population that function produced persists on a different timescale entirely,

carrying capacities that were selected for and are now unwanted, in places built for a purpose that no longer includes them. Institutions have occasionally chosen to meet that situation. The dog statute meets it with adoption, the chimpanzee statute with lifetime care, the horse statute with a legal residency for animals defined precisely by having no owner's mark on them. Each was a decision taken late, after a period in which the default had been disposal or silence.

Skinner's 30 birds held their aim for six years after the equipment was scrapped, and somebody had to decide what to do with a competence that had outlasted its use. That decision is the whole of the matter, and for the street pigeon it has never been taken. The birds on the ledges arrived by domestication and stayed by omission. They are the part of an institution that could not be sold, could not be gathered, and could not be told that its work was over.

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